

AGILE SPRINT PLANNING

Project Title: Customer Support Chatbot for Internal Documentation.

Project Type: RAG-based Web Application.

Developer Name: Pavan Sai Porapu

Domain: AI Automation

Version: v1

Start Date: 02-02-2026

End Date: 31-03-2026

Methodology: Agile

Sprint Duration: 1 week

Total Duration: 8 weeks

Project Status:

Sprint	Start Date	End Date	Status	Outcomes
Sprint 0	02-02-2026	04-02-2026	● Completed	- Project Planning Document - Architecture Diagram
Sprint 1	04-02-2026	10-02-2026	● In progress	- Working User Interface - DB design
Sprint 2	11-02-2026	17-02-2026	○ Not started	- Document Parsing Module - Verified Chunking Output
Sprint 3	18-02-2026	24-02-2026	○ Not started	- VectorDB Setup - Semantic Search
Sprint 4	25-02-2026	03-03-2026	○ Not started	- End to end RAG Pipeline - Result Display
Sprint 5	04-03-2026	10-03-2026	○ Not started	- Confidence Based Routing - Transparent AI decision logic
Sprint 6	11-03-2026	17-03-2026	○ Not started	- Human-in-the-loop system - Self-improving knowledge base
Sprint 7	18-03-2026	24-03-2026	○ Not started	- Admin visibility - Improved response quality
Sprint 8	25-03-2026	31-03-2026	○ Not started	- Stable MVP - Deployment-ready system

Sprint 0

Goal: Reduce risk before coding

Activities:

- Finalize requirements & scope
- Freeze tech stack (React, Django, PostgreSQL, Vector DB)
- Architecture design (RAG pipeline)

Deliverables:

- Architecture diagram
 - Git repository setup
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Sprint 1

Goal: Accept and store user queries.

Tasks:

- Frontend basic UI (query input, submit)
- Django backend setup
- REST API for query submission
- PostgreSQL schema:
 - User Query
 - Query Log

Deliverables:

- Working UI → Backend flow
 - Queries stored in DB
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Sprint 2

Goal: Prepare internal knowledge base

Tasks:

- Document upload API
- Support PDF / DOCX
- Text extraction

- Chunking strategy (overlap + size)
- Metadata handling:
 - doc_id
 - version
 - section

Deliverables:

- Documents successfully parsed and stored
 - Verified chunking output
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Sprint 3

Goal: Enable semantic search

Tasks:

- Embedding generation pipeline
- Vector DB integration (Pinecone / Chroma)
- Store embeddings + metadata
- Similarity search (Top-k)
- Retrieval testing scripts

Deliverables:

- Vector DB populated
 - Semantic retrieval working
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Sprint 4

Goal: Generate answers using retrieved context

Tasks:

- Context construction
- LLM prompt design
- Answer generation
- Source attribution (doc + section)
- End-to-end integration

Deliverables:

- Fully working RAG pipeline
- Answers displayed with sources

Sprint 5

Goal: Decide when AI is confident

Tasks:

- Confidence score components:
- Threshold logic
- High vs Low confidence flow
- UI indicators (confidence badge)

Deliverables:

- Confidence-based routing
 - Transparent AI decision logic
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Sprint 6

Goal: Handle unresolved queries

Tasks

- “Not Resolved” option in UI
- Email escalation service
- Include:
 - User query
 - Retrieved chunks
 - Conversation history
- Agent response storage
- Re-index agent-verified answers

Deliverables:

- Human-in-the-loop system
 - Self-improving knowledge base
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Sprint 7

Goal: Make system observable and maintainable

Tasks:

- Admin dashboard (basic)

- Metrics:
 - Query volume
 - Confidence distribution
 - Escalation rate
- Feedback analytics
- Prompt & retrieval tuning
- Performance optimization

Deliverables:

- Admin visibility
- Improved response quality

Sprint 8

Goal: Production-ready MVP

Tasks:

- End-to-end testing
- Edge-case handling
- Input sanitization
- Authentication (basic)
- Deployment (Docker / Cloud)
- Final documentation:
 - Architecture
 - API docs
 - User guide

Deliverables:

- Stable MVP
- Deployment-ready system
- Final sprint review & retrospective

